Design Proposal of Mach-Zehnder Interferometer

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Abstract: In this course project Mach-Zehnder interferometers are designed and simulations of waveguides and circuits were performed. After fabrication, the measured test results will be compared with the simulation outcomes. The systems were designed for TE 1550nm for circuits fabricated on a 220 nm silicon-on-insulator (SOI) platform.

1. Introduction

Mach-Zehnder interferometers (MZIs) are fundamental building blocks in integrated photonic circuits, enabling optical interference through controlled splitting, propagation, and recombination of guided optical signals. The spectral response of an MZI is determined by the relative phase accumulated between its two arms, which can be controlled through the path-length difference, ΔL. This design proposal focuses on the layout and characterization of waveguides and MZIs operating in cband TE mode at 1550nm. The proposed structures include a symmetric reference MZI and asymmetric MZIs with $\Delta L = 0$, 25, 50, 75, and 100 μm. Here, we use TE 1550nm grating couplers and have variations of the MZIs with Y-branches and directional couplers. In the simulations, different waveguide widths are considered to characterize their influence on waveguide performance.

2. Theory

2.1. Basic Theory of the Photonic Components

2.1.1. Mach–Zehnder Interferometer

In Mach-Zehnder interferometer (MZI) the input optical field is divided by waveguide splitters, resulting in two arms, and recombined at the output. During propagation a relative phase difference is accumulated. In an MZI with an optical path-length difference ΔL, a relative phase difference can be approximated as

$$\Delta\phi = \frac{2\pi n_{\text{eff}} \Delta L}{\lambda}, \tag{1}$$

where n_{eff} is the effective modal index and λ is the operating wavelength. The output transmission varies periodically with wavelength due to constructive and destructive interference. For an asymmetric MZI, the free spectral range (FSR) is approximated with n_g the group index as [1]:

The FSR can be calculated by the expression

$$FSR = \Delta\lambda = \frac{\lambda^2}{\Delta L(n - \lambda\frac{dn}{d\lambda})} = \frac{\lambda^2}{\Delta L n_g} \tag{2}$$

or it can be simplified to just

$$\text{FSR} \approx \frac{\lambda^2}{n_g \Delta L}, \tag{3}$$

3. Modeling and simulations of waveguides

Simulations are performed using Lumerical and INTERCONNECT.

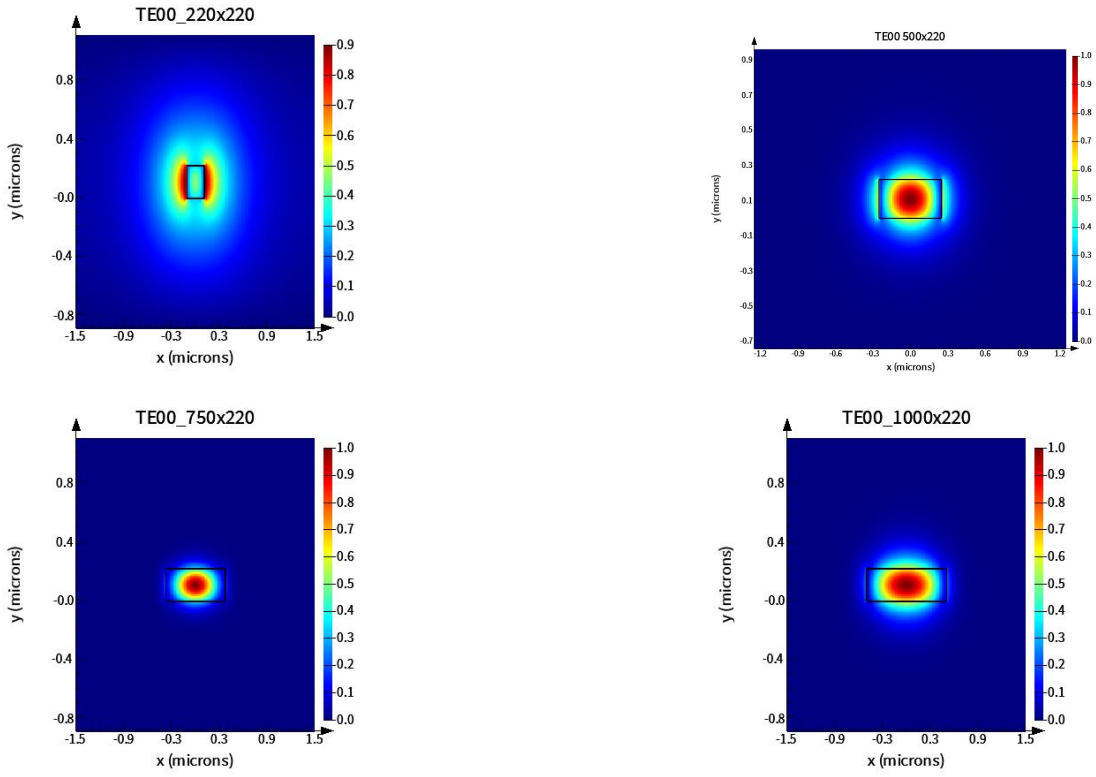


Fig. 1. Different TE00 modes with changing width.

width (microns)	effective index
1000	2.7445
750	2.6662
500	2.4445
220	1.5019

TABLE I
VARIED WIDTHS USED TO SIMULATE THE DIFFERENT TE MODES AND THEIR RESPECTIVE EFFECTIVE INDEX.

In our design, we use strip waveguide with 500nm width and 220nm height. The simulated waveguide mode profiles are presented in the following figures. In this design, the height of the waveguide was set at 220nm, a common value. The waveguide width was varied

The effective index versus wavelength of the simulated waveguide is shown in Figure 7

while the group index versus waveguide is seen in Figure 3.

Using Lumerical the compact model of the waveguide is 2.44435 - 1.13055 * (lambda-1.55) -0.0332067 * (lambda-1.55)^2

this can be shown in 4 below.

A corner analysis was performed on the waveguide. The nominal size of the waveguidde was taken to be 220 nm by 500 nm. however, the wafer was grown to be 219 nm $\pm$ 3.19 nm which gives a bottom error of 215.3 nm and a top error of 223.1 nm. The width variation was taken to be 470 nm to 510 nm to take into account fabrication errors.

Next the simulations of the designed MZI circuits are described are described. The layout and design can be found in in section 4.

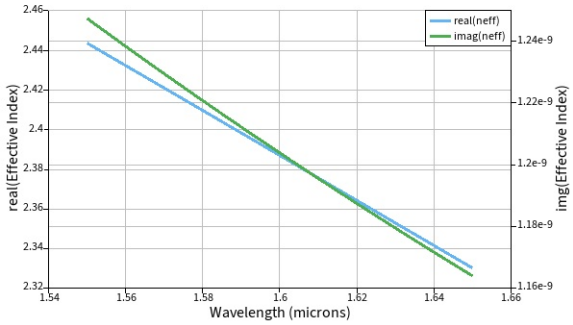


Fig. 2. Effective index versus wavelength.

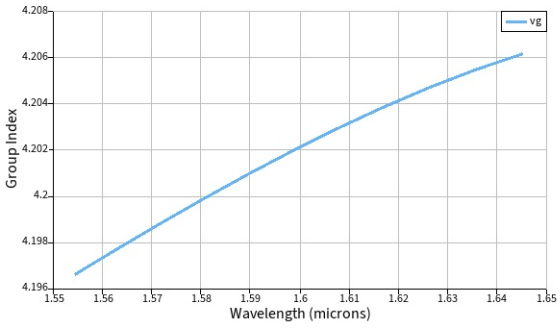


Fig. 3. Group index versus wavelength.

3.1. Simulation of Mach-Zehnder Interferometers

The schematic used to model the Mach-Zehnder Interferometer is shown in

The Free spectral range for changing ΔL is seen in figures 8. The table III compares the FSR or changing ΔL taken from the spectrum with FSR values calculated analytically.

Figure9 shows the transmission vs wavelength simulation obtained from the GitHub design upload. This graph was provided by Lukas Chrostowski.

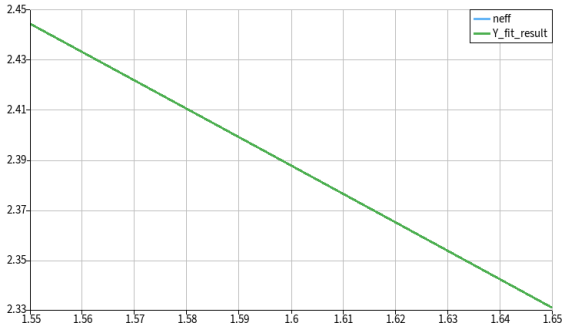


Fig. 4. compact fit model.

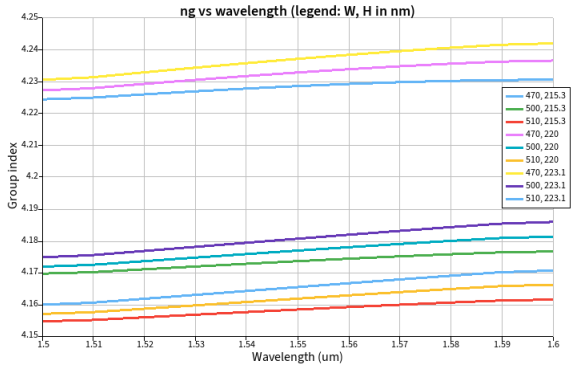


Fig. 5. Group Index vs wavelength for the corner analysis measurements.

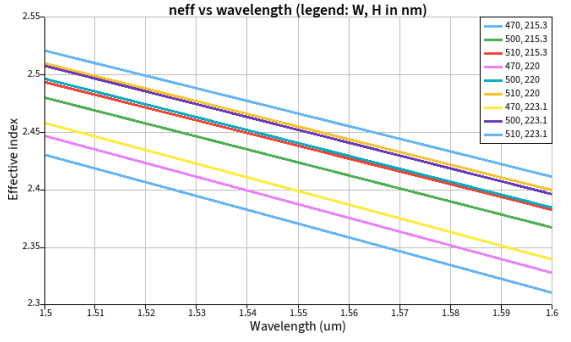


Fig. 6. Effective Index vs wavelength for the corner analysis measurements.

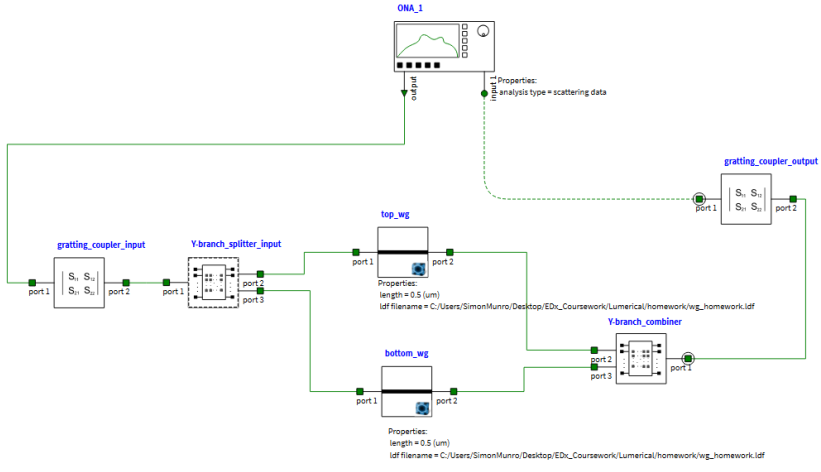


Fig. 7. Diagram of the simulation setup on INTERCONNECT.

ng min	ng max	range
4.14664	4.22391	0.0772731

TABLE II
GROUP INDEX VALUES FROM THE CORNER ANALYSIS TESTS.

ΔL		FSR min (nm)	FSR max (nm)
25	equation	22.7514	23.1754
	spectrum	22.745	23.445
50	equation	11.3757	11.5877
	spectrum	11.42	11.635
75	equation	7.58381	7.72514
	spectrum	7.58	7.735
100	equation	5.68786	5.79385
	spectrum	5.69	5.795

TABLE III
FSR VARIATIONS FOR THE CHANGING ΔL VALUE.

4. Layout

GDSFActory and KLayout were used for the designs of photonic circuits. The proposed layout consists of a systematic set of Mach-Zehnder interferometers (MZI) designed for optical characterization at 1550 nm, see Figure10. The layout includes strip waveguides with a widths of 500 nm together with TE 1550 nm grating couplers (GCs), Y-branches or direct-coupler (DC) structures. A straight loopback between two GCs is added as a reference structure.

- **Grating couplers:** Grating couplers are incorporated for optical input/output coupling at 1550 nm. The pitch between grating couplers belonging to different MZI structures is approximately 200 μm. We are using the $GC_TE_1550_8degOxide_BB$ from the PDK.
- **TE 1550 nm couplers:** Directional coupling structures designed for TE-polarized operation at 1550 nm are included for characterization of the coupling response. We are using the $ebeam_bdc_te1550$ structure from the PDK. We changed the input port of the directional coupler to avoid the risk of the inner one acting as an antenna or reflecting back into the waveguide. Both in and outputs are connected to the left ports of the DC to avoid this.
- **Y-branches:** Y-branches and directional couplers are used to form the interferometric structures to independently evaluate the performance of the corresponding components. We are using $ebeam_y1550$ from the PDK.
- **Mach–Zehnder interferometers:** A series of MZIs with different path-length differences, ΔL, both using Y-branches and directional couplers is included:

  $$\Delta L = 0,\ 25,\ 50,\ 75,\ 100\ \mu\mathrm{m}.$$

  The $\Delta L = 0$ μm structure corresponds to a symmetric MZI as a reference structure, while the remaining structures introduce increasing path-length asymmetry.

Overall, the layout provides a structured test matrix combining different coupling structures, and MZI path-length differences. This enables systematic characterization of the individual building blocks and investigation of the dependence of MZI spectral response on waveguide geometry and path-length difference. Both the waveguide to path and DRC tests in KLayout resulted in no marked errors.

5. Fabrication

The devices are fabricated on a 220 nm silicon-on-insulator (SOI) platform.

Once the design is finalized, electron-beam lithography is used to transfer the pattern onto an electron-beam resist. This resist acts as a protective mask for the waveguides during the etching

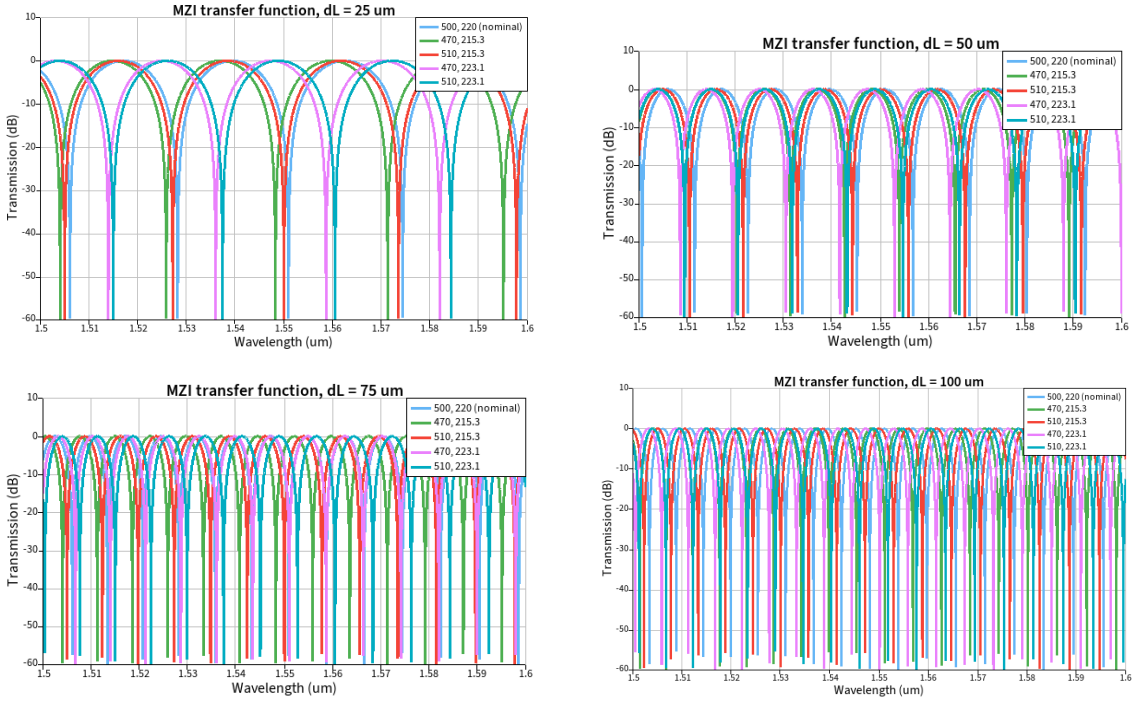


Fig. 8. Different FSR graphs with ΔL ranging from 25 μm to 100 μm.

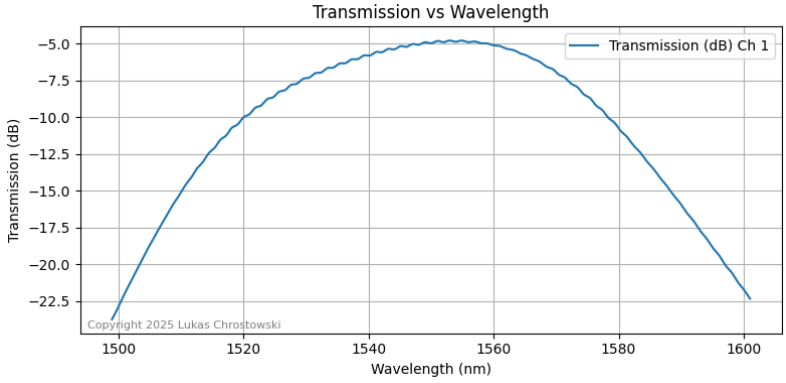


Fig. 9. Simulation of the reference straight waveguide loopback obtain from the GitHub upload simulation from Lukas Chrostowski.

process, ensuring that unwanted silicon is removed while the waveguide structures remain intact.

After the waveguides have been defined through etching, they are covered with an oxide layer to protect the structures and provide optical confinement, completing the fabrication process.

The following are the process definitions for the two fabrication facilities:

Applied Nanotools Inc. NanoSOI Process

Photonic devices were fabricated using Applied Nanotools' NanoSOI multi-project wafer (MPW) process. The devices were patterned on a 220 nm silicon layer over a 2 µm buried oxide (BOX) using 100 keV electron-beam lithography with hydrogen silsesquioxane (HSQ) resist. The patterns were transferred into silicon using chlorine-based inductively coupled plasma reactive-ion etching (ICP-RIE). Finally, a 2.2 µm silicon dioxide cladding layer was deposited by plasma-enhanced

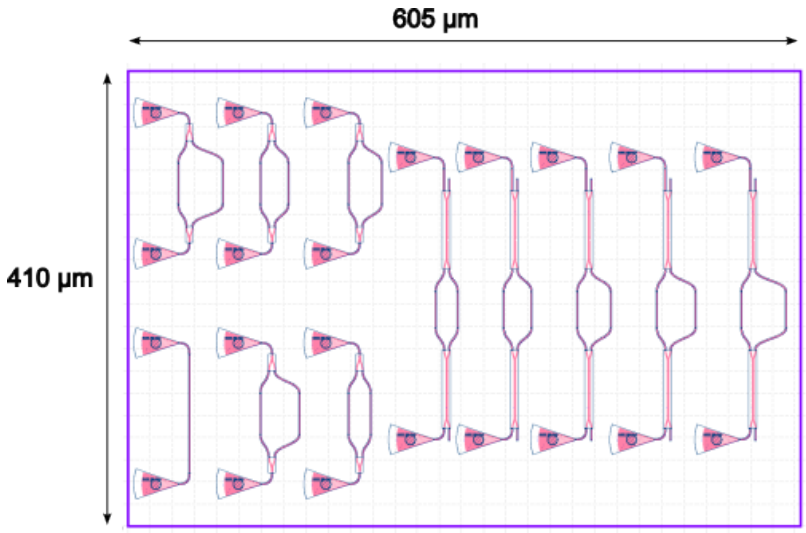


Fig. 10. Design layout of MZIs in KLayout showing the two sets of asymmetric MZIs with either Y-branches or directional couplers as well as a straight loopback.

chemical vapor deposition (PECVD).

Washington Nanofabrication Facility (WNF) Process

Photonic devices were fabricated using 100 keV electron-beam lithography on silicon-on-insulator (SOI) substrates featuring a 220 nm silicon device layer and a 3 µm buried oxide layer. HSQ resist was used to define the patterns, which were transferred into silicon using chlorine-based ICP etching. A silicon dioxide cladding layer was subsequently deposited using PECVD to protect the waveguides and provide optical confinement.

References

References

[1] J. N. Caspers *et al.*, “Mach-zehnder interferometric engineering in silicon optical modulators,” *Applied Optics*, vol. 62, pp. 7387–7397, 2023.